

MARCO FERNSTAEDT

FULL-STACK JAVASCRIPT ENGINEER | APPLICATION MODERNIZATION

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PROFESSIONAL SUMMARY

Full-stack software developer who turns operational pain points into secure, testable software. Experience modernizing workflows with TypeScript, React, Next.js, Node.js, REST APIs, C#.NET, relational data, automated testing, and CI/CD. Strong at extending existing systems incrementally, mapping legacy behavior, protecting working functionality, and shipping accessible operator experiences.

TECHNICAL SKILLS

JavaScript / TypeScript: JavaScript (ES6+), TypeScript, React, Next.js, Node.js, Fastify, Express, Vite

Backend / Data: REST APIs, ASP.NET Core, C#.NET, SQLite, MongoDB, schema migrations, authentication, RBAC

Quality / Delivery: Vitest, Playwright, Testing Library, xUnit, integration testing, GitHub Actions, Git, npm, CI/CD

Platforms / Practices: Windows services, WPF, Linux/VPS, Vercel, PWA, accessibility, secure design, incremental modernization

SELECTED PROJECTS

Cabinet Support Tool | In Development | C#, .NET, ASP.NET Core, WPF, SQLite

- Developing a general support tool to monitor and manage multiple Windows machines through a Windows service agent, central API, and desktop console.
- Working on secure device enrollment, approved access, heartbeat reporting, process and event diagnostics, and connectivity monitoring over outbound HTTPS.
- Extending the existing architecture incrementally, with typed commands, bounded results, audit records, and automated regression tests.
- Planning desired-state configuration and firewall drift detection with policy-approved remediation, connectivity verification, and rollback. Production deployment and native Windows acceptance remain pending.

Household Coordinator PWA | TypeScript, React, Fastify, Node.js, SQLite, Vite

- Built a production-minded, installable PWA with server-enforced tenant isolation, owner/member authorization, expiring scoped guest access, optimistic concurrency, replay-safe mutations, and truthful offline conflict recovery.
- Implemented schema-validated APIs, Argon2id authentication, HttpOnly sessions, CSRF/origin checks, rate limiting, migrations, and automated unit, accessibility, browser, and offline tests with Vitest and Playwright.

Accessible Image-to-Audio Platform | Next.js, React, TypeScript, REST APIs, Vercel

- Developed and deployed a full-stack application that validates uploaded images, generates descriptions and spoken audio through server-side APIs, and returns binary audio for accessible playback and download.
- Added keyboard and screen-reader workflows, semantic status reporting, server-side validation, rate limiting, bot protection, secret isolation, CI checks, and production-build verification.

ENGINEERING APPROACH

Incremental modernization | Legacy-system discovery | API and data-model design | Secure-by-default implementation | Test-driven delivery | Accessible UI engineering | Operational documentation | Cross-functional problem solving